

Zephyr 應用程式dm9051網路實作案例 - dhcpv4_client

系統版本環境：nRF and Zephyr

Booting nRF Connect SDK v3.1.0-a7a6d338252f

Using Zephyr OS v4.1.99-5f4c874a5ee8

前言：dhcpv4_client應用的 .conf 與.overlay套用關係

- Zephyr 構建時會依序合併：板級 *_defconfig → 應用層 prj.conf → 額外覆寫的 overlay 檔。
- 本範例的核心設定在 prj.conf，提供共用的網路/協定能力；若需硬體或情境差異，則以 overlay 疊加。
- 主要 config：overlay_nrf54l15.conf 負責 nRF54L15專有的配置，overlay_dm9051.conf負責 DM9051 的網路驅動與底層緩衝配置。
- 建議在指令加入 -DOVERLAY_CONFIG="boards/overlay_nrf54l15.conf" 這類串列，確保 base (prj.conf) 先生效，再讓 overlay 逐一覆寫細節。
- 指令加入 -DOVERLAY_CONFIG="boards/overlay_nrf54l15.conf;overlay_dm9051.conf"，可配置多個overlay配置，以分號分隔。
- 主要 overlay：overlay_nrf54l15.overlay 負責 nRF54L15專有的設定，'overlay_dm9051.overlay' 負責 DM9051 的網路驅動與底層硬件設定。

建置命令範例

```
# 使用基礎配置
west build -b nrf54l15dk/nrf54l15/cpuapp

# 使用 overlay 配置
west build -b nrf54l15dk/nrf54l15/cpuapp -- -
DOVERLAY_CONFIG="boards/overlay_nrf54l15.conf"

# 使用多個 overlay 配置 (用分號分隔)
west build -b nrf54l15dk/nrf54l15/cpuapp -- -
DOVERLAY_CONFIG="boards/overlay_nrf54l15.conf;overlay_dm9051.conf";
```

prj.conf

```
CONFIG_NET_DHCPV4=y
CONFIG_NET_DHCPV4_OPTION_CALLBACKS=y
CONFIG_DNS_RESOLVER=y
```

```
CONFIG_INIT_STACKS=y
```

```
CONFIG_NET_MGMT=y
```

```
CONFIG_NET_MGMT_EVENT=y
```

```
CONFIG_LOG=y
```

```
CONFIG_NET_SHELL=y
```

boards/overlay_nrf54l15.conf

```
CONFIG_CLOCK_CONTROL_NRF_K32SRC_RC=y
```

```
CONFIG_LOG_BACKEND_UART=y
```

```
CONFIG_SHELL_LOG_BACKEND=n
```

overlay_dm9051.conf

```
CONFIG_NETWORKING=y
```

```
CONFIG_NET_IPV6=n
```

```
CONFIG_NET_IPV4=y
```

```
CONFIG_NET_ARP=y
```

```
CONFIG_NET_UDP=y
```

```
CONFIG_NET_L2_ETHERNET=y
```

```
CONFIG_NET_BUF_DATA_SIZE=2048
```

```
CONFIG_NET_IPV4_FRAGMENT=y
```

```
CONFIG_ETH_DM9051=y
```

```
CONFIG_SPI=y
```

```
CONFIG_GPIO=y
```

boards/overlay_nrf54l15.overlay

```
&rram_controller {
```

```
    /delete-node/ cpufldr_sram;
```

```
    /delete-node/ cpufldr_rram;
```

```
};
```

```

/* Adjust the cpuapp_sram to include the freed up cpuflpr_sram */

&cpuapp_sram {

    reg = <0x20000000 DT_SIZE_K(256)>;

    ranges = <0x0 0x20000000 0x20040000>;

};


/* Adjust the cpuapp_rram to include the freed up cpuflpr_rram */

&cpuapp_rram {

    reg = <0x0 DT_SIZE_K(1524)>;

};


/* Disable NFCT to free up NFC pins for GPIO usage */

&nfct {

    status = "disabled";

};


/* Configure UICR to use NFCT pins as GPIOs */

&uicr {

    nfct-pins-as-gpios;

};

```

overlay_dm9051.overlay

```

/* SPI 腳位在 gpio1，必須啟用 gpiote20 */
&gpio1 {
    status = "okay";
};

&gpiote20 {
    status = "okay";
};

&pinctrl {
    spi21_default: spi21_default {
        group1 {
            psels = <NRF_PSEL(SPIM_SCK, 1, 3)>,
                <NRF_PSEL(SPIM_MOSI, 1, 1)>,
                <NRF_PSEL(SPIM_MISO, 1, 0)>;
        };
    };
};

&spi21 {
    status = "okay";
    cs-gpios = <&gpio1 2 GPIO_ACTIVE_LOW>; /* CS on P1.2 */
    pinctrl-0 = <&spi21_default>;
    pinctrl-names = "default";

    dm9051: dm9051@0 {
        compatible = "davicom,dm9051";
        reg = <0>;
        spi-max-frequency = <1000000>; /* 1MHz - minimum for debugging */
        /* INT 腳位 */
        /* int-gpios = <&gpio0 3 GPIO_ACTIVE_LOW>; */
        /* Reset 腳位 */
        /* reset-gpios = <&gpio2 0 GPIO_ACTIVE_LOW>; */
        local-mac-address = [00 00 00 00 00 00]; /* [00 11 22 33 44 55]; */
    };
};

```

專案目錄結構圖

```

zephyr/
├── dts/bindings/ethernet/
│   └── davicom,dm9051.yaml
│
├── samples/net/dhcpv4_client/
│   └── CMakeLists.txt
└── # 建置腳本

```

```
|   ├── prj.conf                # 主配置檔案
|   ├── boards/
|   |   ├── overlay_nrf54l15.conf    # 板級配置
|   |   └── overlay_nrf54l15.overlay  # 板級 Device Tree
|   ├── overlay_dm9051.conf          # DM9051配置檔案
|   ├── overlay_dm9051.overlay       # DM9051 Device Tree
|   └── src/
|       └── main.c                  # 應用程式主程式
└── drivers/ethernet/
    ├── CMakeLists.txt             # 建置腳本
    ├── Kconfig                    # Kconfig
    ├── Kconfig.dm9051             # DM9051 Kconfig(配置選項)
    ├── eth_dm9051.c                # DM9051 驅動實作
    └── eth_dm9051_priv.h           # DM9051 私有標頭檔
```